Cambridge (CIE) A Level Maths: Pure 3

Your notes

Complex Numbers

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Introduction to complex numbers

Complex numbers are a set of numbers which contain both a real part and an imaginary part. The set of complex numbers is denoted as $\mathbb{C}$.

What is an imaginary number?

- Up until now, when we have encountered an equation such as $x^2 = -1$ we would have stated that there are "no real solutions" as the solutions are $x = \pm\sqrt{-1}$ which are not real numbers
- To solve this issue, mathematicians have defined one of the square roots of negative one as i; an imaginary number
  - $\sqrt{-1} = \mathrm{i}$
  - $\mathrm{i}^2 = -1$
- We can use the rules for manipulating surds to manipulate imaginary numbers.
- We can do this by rewriting surds to be a multiple of $\sqrt{-1}$ using the fact that $\sqrt{ab} = \sqrt{a} \times \sqrt{b}$

What is a complex number?

- Complex numbers have both a real part and an imaginary part
  - For example: $3 + 4\mathrm{i}$
  - The real part is 3 and the imaginary part is 4
    - Note that the imaginary part does not include the 'i'
- Complex numbers are often denoted by z and we can refer to the real and imaginary parts respectively using $\mathrm{Re}(z)$ and $\mathrm{Im}(z)$
- In general:
  - $z = a + b\mathrm{i}$ This is the Cartesian form of z
  - $\mathrm{Re}(z) = a$
  - $\mathrm{Im}(z) = b$
- It is important to note that two complex numbers are equal if, and only if, both the real and imaginary parts are identical.
  - For example, $3 + 2\mathrm{i}$ and $3 + 3\mathrm{i}$ are **not equal**

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Worked Example

(a) Solve the equation $x^2 = -9$

(b) Solve the equation $(x + 7)^2 = -16$.

a) $x^2 = -9$

$x = \pm\sqrt{-9}$

USING $\sqrt{ab} = \sqrt{a} \times \sqrt{b}$: $x = \pm\sqrt{9}\sqrt{-1}$

$x = \pm 3i$

b) $(x + 7)^2 = -16$

$x + 7 = \pm\sqrt{-16}$

$x + 7 = \pm\sqrt{16}\sqrt{-1}$

$x + 7 = \pm 4i$

$x = -7 \pm 4i$

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Examiner Tips and Tricks

- Be careful in your notation of complex and imaginary numbers.
- For example:
  $(3\sqrt{5})i$ could also be written as $3i\sqrt{5}$, but if you wrote $3\sqrt{5}\,i$ this could easily be confused with $3\sqrt{5i}$.

Operations with complex numbers

How do I add and subtract complex numbers?

- When adding and subtracting complex numbers, simplify the real and imaginary parts separately
  - Just like you would when collecting like terms in algebra and surds, or dealing with different components in vectors
  - $(a + b\mathrm{i}) + (c + d\mathrm{i}) = (a + c) + (b + d)\mathrm{i}$
- Complex numbers can also be multiplied by a constant in the same way as algebraic expressions:
  - $k(a + b\mathrm{i}) = ka + kb\mathrm{i}$

How do I multiply complex numbers?

- The most important thing to bear in mind when multiplying complex numbers is that $i^2 = -1$
- We can still apply our usual rules for multiplying algebraic terms:
  - $a(b+c) = ab + ac$
  - $(a+b)(c+d) = ac + ad + bc + bd$
- Sometimes when a question describes multiple complex numbers, the notation $z_1, z_2, \ldots$ is used to represent each complex number

What are the powers of i?

- Because $i^2 = -1$ this can lead to some interesting results for higher powers of i
  - $i^3 = i^2 \times i = -i$
  - $i^4 = (i^2)^2 = (-1)^2 = 1$
  - $i^5 = (i^2)^2 \times i = i$
  - $i^6 = (i^2)^3 = (-1)^3 = -1$
- We can use this same approach of using i^2 to deal with much higher powers
  - $i^{23} = (i^2)^{11} \times i = (-1)^{11} \times i = -i$
  - Just remember that -1 raised to an even power is 1 and raised to an odd power is -1

Worked Example

(a) Simplify the expression $2(8 - 6i) - 5(3 + 4i)$.

(b) Given two complex numbers $z_1 = 3 + 4i$ and $z_2 = 6 + 7i$. Find $z_1 \times z_2$.

a) EXPANDING: $16 - 12i - (15 + 20i)$

COLLECTING THE REAL AND IMAGINARY PARTS: $16 - 15 - 12i - 20i$

SIMPLIFY: $1 - 32i$

b) EXPAND USING PREFERRED METHOD: $z_1 \times z_2 = 18 + 21i + 24i + 28i^2$

COLLECT LIKE TERMS: $18 + 45i + 28i^2$

RECALL $i^2 = -1$: $18 + 45i + 28(-1)$

SIMPLIFY: $-10 + 45i$

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Examiner Tips and Tricks

- Most calculators used at A-Level can work with complex numbers and you can use these to check your working.
- You should still show your full working though to ensure you get all marks though.

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Complex conjugation & division

When dividing **complex** numbers, we can use the **complex conjugate** to make the denominator a real number, which makes carrying out the division much easier.

What is a complex conjugate?

- For a given complex number $z = a + bi$, **the complex conjugate of z is denoted as z^***, where $z^* = a - bi$
- If $z = a - bi$ then $z^* = a + bi$
- You will find that:
  - $z + z^*$ is always real because $(a + bi) + (a - bi) = 2a$
    - For example: $(6 + 5i) + (6 - 5i) = 6 + 6 + 5i - 5i = 12$
  - $z - z^*$ is always imaginary because $(a + bi) - (a - bi) = 2bi$
    - For example: $(6 + 5i) - (6 - 5i) = 6 - 6 + 5i - (-5i) = 10i$
  - $z \times z^*$ is always real because $(a + bi)(a - bi) = a^2 + abi - abi - b^2i^2 = a^2 + b^2$ (as $i^2 = -1$)
    - For example: $(6 + 5i)(6 - 5i) = 36 + 30i - 30i - 25i^2 = 36 - 25(-1) = 61$

How do I divide complex numbers?

- When we divide complex numbers, we can express the calculation in the form of a fraction, and then **start by multiplying the top and bottom by the conjugate of the denominator**:
  - $$\frac{a + bi}{c + di} = \frac{a + bi}{c + di} \times \frac{c - di}{c - di}$$
- This ensures we are multiplying by 1; so not affecting the overall value
- This gives us a real number as the denominator because we have a complex number multiplied by its conjugate (zz^*)
- This process is very similar to "rationalising the denominator" with surds

Worked Example

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Find the value of $(1 + 7i) \div (3 - i)$.

$\frac{1+7i}{3-i}$ — REWRITE AS FRACTION

$\frac{1+7i}{3-i} \times \frac{3+i}{3+i}$ — MULTIPLY TOP AND BOTTOM BY CONJUGATE OF BOTTOM. NOTICE THAT $\frac{3+i}{3+i} = 1$ SO OVERALL VALUE IS NOT AFFECTED.

$\frac{3+i+21i+7i^2}{9+3i-3i-i^2}$ — MULTIPLY OUT. NOTICE THE IMAGINARY PARTS ON THE BOTTOM ELIMINATE EACH OTHER. CAREFUL: $-i \times i = -i^2$

$\frac{3+22i-7}{9+1}$ — SIMPLIFY, RECALLING $i^2 = -1$

$\frac{-4+22i}{10}$ — WRITE AS TWO TERMS, SIMPLIFYING FRACTIONS.

$= -\frac{4}{10} + \frac{22}{10}i = -\frac{2}{5} + \frac{11}{5}i$

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Examiner Tips and Tricks

- We can speed up the process for finding zz^* by using the basic pattern of $(x+a)(x-a) = x^2 - a^2$
- We can apply this to complex numbers: $(a+bi)(a-bi) = a^2 - b^2i^2 = a^2 + b^2$ (using the fact that $i^2 = -1$)
- So $3+4i$ multiplied by its conjugate would be $3^2 + 4^2 = 25$

Your notes

Square roots of a complex number

How do I find the square root of a complex number?

- The square roots of a complex number will themselves be complex:
  - i.e. if $z^2 = a + b\mathrm{i}$ then $z = c + d\mathrm{i}$
- We can then square $(c + d\mathrm{i})$ and equate it to the original complex number $(a + b\mathrm{i})$, as they both describe z^2:
  - $a + b\mathrm{i} = (c + d\mathrm{i})^2$
- Then expand and simplify:
  - $a + b\mathrm{i} = c^2 + 2cd\mathrm{i} + d^2\mathrm{i}^2$
  - $a + b\mathrm{i} = c^2 + 2cd\mathrm{i} - d^2$
- As both sides are equal we are able to equate real and imaginary parts:
  - Equating the real components: $a = c^2 - d^2$ (1)
  - Equating the imaginary components: $b = 2cd$ (2)
- These equations can then be solved simultaneously to find the real and imaginary components of the square root
  - In general, we can rearrange (2) to make $\frac{b}{2d} = c$ and then substitute into (1)
  - This will lead to a **quartic** equation in terms of d; which can be solved by making a substitution to turn it into a quadratic (see 1.1.5 Further Solving Quadratic Equations (Hidden Quadratics))
- The values of d can then be used to find the corresponding values of c, so we now have both components of both square roots $(c + d\mathrm{i})$
- Note that one root will be the negative of the other root
  - i.e. $c + d\mathrm{i}$ and $-c - d\mathrm{i}$

Worked Example

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Find the square roots of 5+12i.

Let $z^2 = 5 + 12i$ and $z = a + bi$

ASSUME THAT THE SQUARE ROOT OF THE NUMBER IS COMPLEX

$z^2 = a^2 + 2abi + b^2i^2$
$z^2 = a^2 + 2abi - b^2$

SQUARE THE ROOT, TO GIVE YOU ANOTHER EXPRESSION FOR z^2

$5 + 12i = a^2 + 2abi - b^2$

EQUATE THE TWO EXPRESSIONS, WHICH ARE BOTH DESCRIBING z^2

REAL COMPONENTS | IMAGINARY COMPONENTS

(1) $5 = a^2 - b^2$ (2) $12 = 2ab$

THE TWO SIDES OF THE EQUATION ARE THE SAME, SO THE REAL COMPONENTS OF BOTH SIDES MUST BE EQUAL, AS MUST BE THE IMAGINARY COMPONENTS

REARRANGING (2) $\frac{6}{b} = a$

SUBSTITUTING INTO (1) $5 = \left(\frac{6}{b}\right)^2 - b^2$

$5 = \frac{36}{b^2} - b^2$

MULTIPLY BY b^2 $5b^2 = 36 - b^4$

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THIS PRODUCES A QUARTIC, BUT YOU MAY WISH TO CONSIDER A QUADRATIC OF THE FORM $y^2 + 5y - 36 = 0$ TO HELP SOLVE IT

$b^4 + 5b^2 - 36 = 0$
$(b^2 + 9)(b^2 - 4) = 0$
$b^2 = -9$ $b^2 = 4$
$b = \pm 2$

No real solutions

BECAUSE WE PREVIOUSLY STATED z=a+bi, b MUST BE A REAL NUMBER

(2) $a = \frac{6}{b}$

$a = \frac{6}{2}$ or $a = \frac{6}{-2}$

$a = 3$ or $a = -3$

SUBSTITUTE THE POSSIBLE VALUES OF b BACK INTO (2) TO FIND THE POSSIBLE VALUES FOR a

SO $z = 3 + 2i$ OR $z = -3 - 2i$ ARE THE SQUARE ROOTS OF $5 + 12i$

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Examiner Tips and Tricks

- Most calculators used at A-Level can handle complex numbers.
- Once you have found the square roots algebraically; use your calculator to square them and make sure you get the number you were originally trying to square-root!

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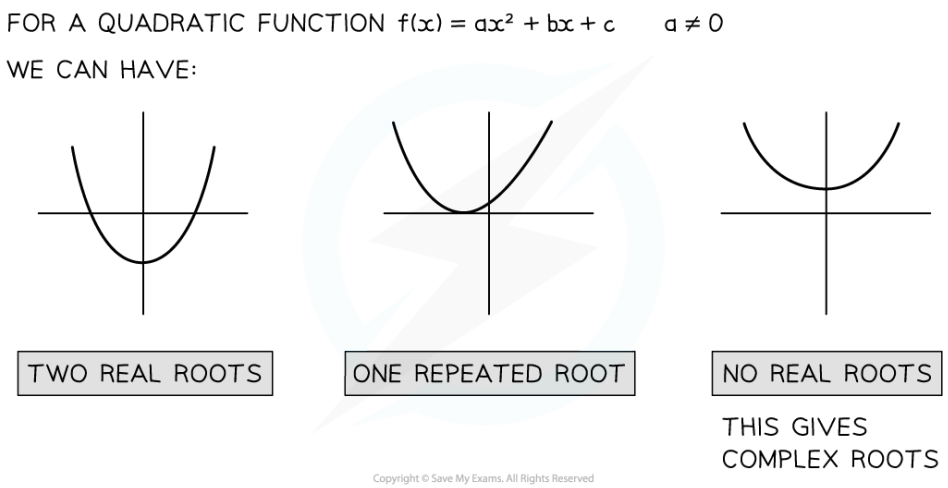
Complex roots of quadratics

What are complex roots?

- **Complex numbers** provide solutions for quadratic equations which have **no real roots**



- Complex roots occur when solving a quadratic with a **negative discriminant**
  - This leads to square rooting a negative number

How do I solve a quadratic equation with complex roots?

- We solve an equation with complex roots in the same way we solve any other quadratic equations
  - If in the form $az^2 + b = 0 \ (a \neq 0)$ we can **rearrange** to solve
  - If in the form $az^2 + bz + c = 0 \ (a \neq 0)$ we can **complete the square** or use the **quadratic formula**
- We use the property $\mathrm{i} = \sqrt{-1}$ along with a manipulation of surds
- $\sqrt{-a} = \sqrt{a \times -1} = \sqrt{a} \times \sqrt{-1}$
- When the **coefficients** of the quadratic equation are **real**, complex roots occur in **complex conjugate pairs**
  - If $z = m + n\mathrm{i} \ (\mathrm{n} \neq 0)$ is a root of a quadratic with real coefficients then $z^* = m - n\mathrm{i}$ is also a root
- When the coefficients of the quadratic equation are **non-real**, the solutions will **not** be complex conjugates

- To solve these use the quadratic formula

Your notes

How do I find a quadratic equation given one complex root?

- We can find the equation of the form $z^2 + bz + c = 0$ if you are given a complex root in the form $m + ni$
  - We know that the complex conjugate $m - ni$ is another root,
  - This means that $z - (m + ni)$ and $z - (m - ni)$ are **factors** of the quadratic equation
  - Therefore $z^2 + bz + c = [z - (m + ni)][z - (m - ni)]$
    - Writing this as the difference of two squares - $((z - m) - ni)((z - m) + ni)$ - will speed up expanding
  - **Expanding** and **simplifying** gives us a quadratic equation where b and c are **real** numbers

Worked Example

Solve the following for z, giving your answers in the form $a + bi$ where a and b are real numbers.

(i) $z^2 + 25 = 0$

(ii) $z^2 - 2z + 5 = 0$

In each case, be sure to show your working.

(i) $z^2 + 25 = 0$ REARRANGE FOR z

$z^2 = -25$

$z = \pm\sqrt{-25}$ USING $\sqrt{ab} = \sqrt{a} \times \sqrt{b}$ AND $i = \sqrt{-1}$

$\sqrt{-25} = \sqrt{25 \times -1}$

$= \sqrt{25} \times \sqrt{-1}$

$= 5i$

$z = \pm 5i$

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(ii) $z^2 - 2z + 5 = 0$

USING THE QUADRATIC FORMULA

$a = 1 \quad b = -2 \quad c = 5$

$$z = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

$$z = \frac{-(-2) \pm \sqrt{(-2)^2 - 4(1)(5)}}{2(1)}$$

$$z = \frac{2 \pm \sqrt{-16}}{2}$$

SIMPLIFY

$$z = \frac{2 \pm \sqrt{16}\sqrt{-1}}{2}$$

SPLIT THE NEGATIVE ROOT AND USE i

$$z = \frac{2 \pm 4i}{2}$$

SIMPLIFY

$z = 1 \pm 2i$

ALTERNATIVE METHOD – COMPLETING THE SQUARE

$z^2 - 2z + 5 = 0$

$(z - 1)^2 - 1^2 + 5 = 0$

$(z - 1)^2 + 4 = 0$

$(z - 1)^2 = -4$

$z - 1 = \pm\sqrt{-4}$

$z - 1 = \pm 2i$

$z = 1 \pm 2i$

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Examiner Tips and Tricks

- Once you have your final answers you can check your roots are correct by substituting your solutions back into the original equation.
- You should get 0 if correct! [Note: 0 is equivalent to $0 + 0\mathbf{i}$]

Complex Roots of cubics & quartics

How many roots should a polynomial have?

- We know from previous sections that every **quadratic** equation has **two roots** (not necessarily distinct)
- This is a particular case of a more general rule:
  - **Every polynomial equation, with real coefficients, of degree *n* has *n* roots**
  - The *n* roots are not necessarily all **distinct** and therefore we need to count any **repeated** roots that may occur individually
- From the above rule we can state the following:
  - A **cubic** equation of the form $ax^3 + bx^2 + cx + d = 0$ can have either:
    - 3 **real** roots
    - Or 1 **real** root and a complex **conjugate** pair
- A **quartic** equation of the form $ax^4 + bx^3 + cx^2 + dx + e = 0$ will have the following cases for roots:

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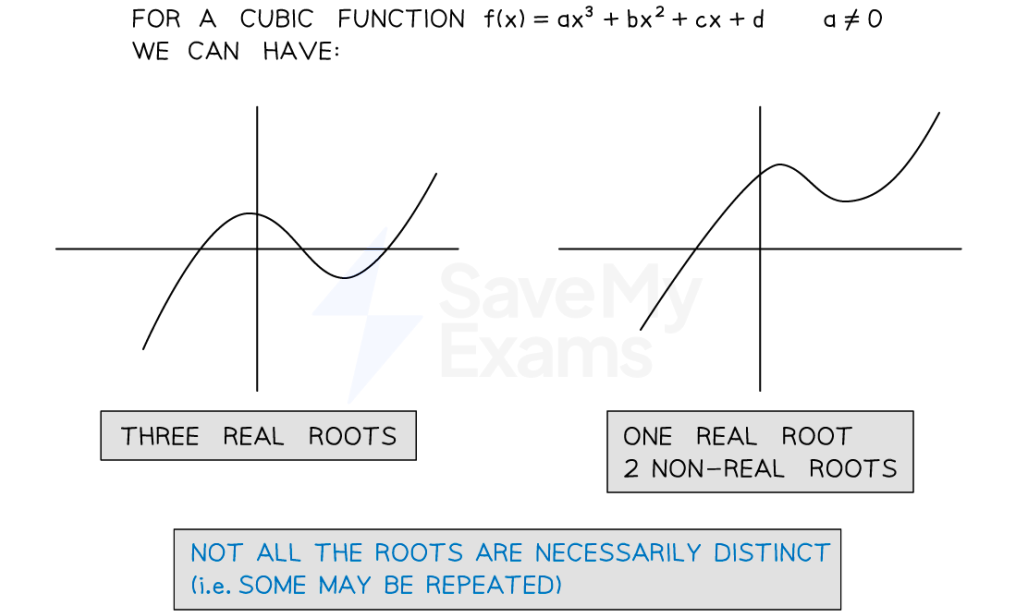
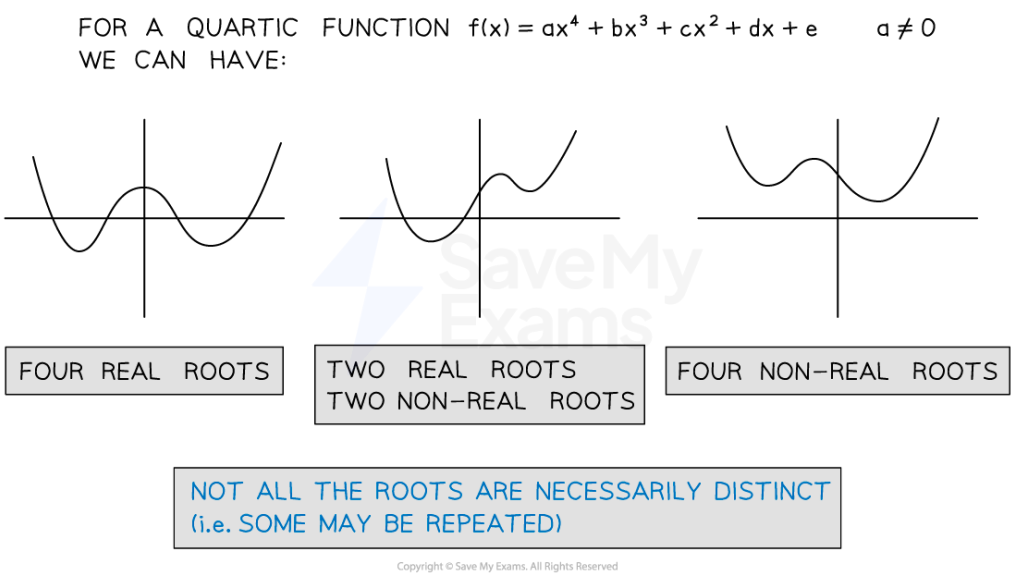
- 4 **real** roots
- 2 **real** and 2 **non-real** roots(a complex conjugate pair)
- 4 **non-real** roots (two complex conjugate pairs)



Number of roots of a cubic function



Number of roots of a quartic function

How do I solve a cubic equation given one complex root?

- Steps to solve a cubic equation with complex roots

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Your notes

- If we are told that $m + ni$ is a root, then we know $m - ni$ is also a root
  - This means that $(z - (m + ni))$ and $(z - (m - ni))$ are **factors** of the cubic equation
  - **Multiply** the above factors together gives us a quadratic factor of the form $(Az^2 + Bz + C)$
  - We need to find the third factor $(z - \alpha)$
  - **Multiply** the factors and **equate** to our original equation to get
    $$(Az^2 + Bz + C)(z - \alpha) = ax^3 + bx^2 + cx + d$$
- From there either
  - **Expand** and **compare coefficients** to find α
  - Or use **polynomial division** to find the factor $(z - \alpha)$
- Finally, write your **three roots** clearly

How do I solve a quartic equation given one complex root?

- When asked to find the roots of a quartic equation when we are given one, we use almost the same method as we did for a cubic equation
  - State the initial root and its conjugate and write their factors as a quadratic factor (as above) we will have two unknown roots to find, write these as factors $(z - \alpha)$ and $(z - \beta)$
  - The unknown factors also form a quadratic factor $(z - \alpha)(z - \beta)$
  - Then continue with the steps from above, either **comparing coefficients** or using **polynomial division**
    - If using polynomial division, then solve the quadratic factor you get to find the roots α and β

How do I find unknown coefficients of cubics and quartics?

- Steps to find unknown variables in a given equation when given a root:
  - **Substitute** the given root into the equation $f(z) = 0$
  - **Expand** and **group** together the **real** and **imaginary** parts (these expressions will contain our unknown values)
  - **Solve** as simultaneous equations to find the unknowns
  - **Substitute** the values into the **original** equation

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- From here continue using the previously described methods for finding other roots for either cubic/quartic equations

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Worked Example

Given that $2 - 3i$ is a root of the cubic equation $z^3 - 6z^2 + 21z - 26 = 0$, find the other roots.

$z^3 - 6z^2 + 21z - 26 = 0$

GIVEN ROOT $z = 2 - 3i$

CONJUGATE ROOT $z = 2 + 3i$

WRITE THE THIRD ROOT AS $z = \alpha$ (WHICH IS TO BE FOUND)

WRITE THE ROOTS AS FACTORS AND EQUATE TO ORIGINAL EQUATION

$(z - (2 - 3i)(z - (2 + 3i))(z - \alpha) \equiv z^3 - 6z^2 + 21z - 26$

EXPAND AND SIMPLIFY THE FIRST TWO BRACKETS

$(z - (2 - 3i))(z - (2 + 3i))$

$(z - 2 + 3i)(z - 2 - 3i)$

$z^2 - 2z - 3zi - 2z + 4 + 6i + 3zi - 6i - 9i^2$

SIMPLIFY LIKE TERMS

$z^2 - 4z + 13$

USE $i^2 = -1$

THEREFORE

$(z^2 - 4z + 13)(z - \alpha) = z^3 - 6z^2 + 21z - 26$

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FROM HERE WE HAVE TWO OPTIONS

METHOD 1 – EXPAND AND COMPARE COEFFICIENTS TO FIND α

$z^3 - 4z^2 + 13z - \alpha z^2 + 4\alpha z - 13\alpha = z^3 - 6z^2 + 21z - 26$

$z^3 + (-4 - \alpha)z^2 + (13 + 4\alpha)z - 13\alpha = z^3 - 6z^2 + 21z - 26$

COMPARING THE CONSTANT TERM

$-13\alpha = -26$

$\alpha = 2$

CHECK USING COEFFICIENTS OF z

$13 + 4\alpha = 21$

$4\alpha = 8$

$\alpha = 2$

$z = 2$ IS A ROOT

OUR ROOTS ARE

$2 + 3i,\ 2 - 3i,\ 2$

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Examiner Tips and Tricks

- As with solving quadratic equations, we can substitute our solutions back into the original equation to check we get 0.